

DBbun Invisible Illness Burden Simulator: quality assurance log

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How quality was checked

Check	Who or what did it	What it found	What changed
Automated tests	Python test suite (unittest)	227 tests pass, 5 skipped.	Every defect fixed gets a regression test.
Own five-part audit	Claude Code with Claude Sonnet 5.5; read-only first, reports kept	Accuracy, function, federal-resource, interface and wording problems.	Fixed in batches, then re-tested.
AI model critiques of earlier versions of every simulator PDF (25) and comparison PDF (13)	Claude Sonnet 5.5, Grok 4.5, Gemini 3.5 (Google), GPT (OpenAI; exact build not exposed) and GPT-5 Codex, given one rubric-based prompt	Reviewers rated the comparison PDFs 3 to 4 of 5 and raised the same themes across simulators: price-year and currency labels, non-U.S. data compared with U.S. figures, text that broke the PDF layout, over-confident model wording.	Fixed once in the code and prompts, not simulator by simulator. All 25 simulators were then regenerated and the 13 comparisons rebuilt (about \$21 in model calls).
AI code review	GPT-5 Codex; report only, it changed no code	Ten findings.	Nine fixed with regression tests; one documented (the browser token is a single-user limit).
Federal cross-checks	Code, not the AI model	Each figure a paper states is restated in current prices or compared with CDC and survey data.	Mismatches are shown on page 1 of each PDF, not hidden.
FHIR export	Official HL7 FHIR validator, R4 4.0.1, terminology checking off	26 files (25 simulators and one reuse demonstration; the second reuse demonstration was not validated): 0 errors. 40 to 74 best-practice warnings each: no patient, no performer or time, no narrative text, DBbun-defined codes.	First run gave 28 errors per file (undefined DBbun extensions); fixed by shipping their definitions with each bundle.
Public demo	Rebuilt from the same code and opened live	25 simulators, 13 comparisons and the third-condition demo load; FAQ and downloads match the local tool.	Rebuilt with the audio overviews, simulator numbering, the feedback log and the accuracy fixes.
Generated Python check	check_generated_python.py, run locally 10 October 2026	The Python files in all 25 library bundles reproduced the app's results (605 values: defaults, every scenario, forecasts).	New simulators' files are not run automatically.
Second AI review of the whole package	GPT (OpenAI; exact build not exposed), 10 October 2026; report only	Excel holds saved values, not formulas; one comparison said figures were "not provided" while citing them; some cost ranges and PDF text were stale.	Wording corrected, that comparison regenerated with a code check, stale text fixed.

How reviewer advice was handled

I fixed only problems that were real and systematic, checking each against the paper or the data first. Wording that claimed more than the evidence showed was not used. The reviewers are tools, not authorities: I decided what to adopt.

What this log does not show

- The outside reviews covered the earlier versions. The regenerated simulators were checked by the tests, the cross-checks and my own spot checks, without a second full outside review.
- No independent person has yet completed a documented, uncoached task in the public demo. Two outside readers (a neighborhood-group member and a person with lived experience of Lyme disease and a research background) gave written feedback without doing the tasks: the first screen was too dense, the page text-heavy and my time estimate too short. I added a skip-the-grid note, a collapsed coverage map, one-line study summaries, a Lyme / Long COVID filter, a "what this is and is not" banner and FAQ pictures, and shortened my request. Further results will be added here only if they happen.
- The 38 audio overviews are read by a synthetic voice from scripts composed from each saved record, with a source table mapping paragraphs to the document; three scripts were also reviewed paragraph by paragraph with Claude Code, the rest by automated rules only. The FHIR export is a prototype: not clinical data, not profiled to US Core or USCDI+, not certified; terminology checking was off.
- The two demonstrations on other conditions (a paper on post-surgical chronic pain and an emergency-care cost paper) used the general pipeline only. The cost paper's dollar amounts were restated in current prices by the price-index check; neither has curated federal evidence for its condition or outside validation.

AI tools and model IDs

Inside the product: Claude Sonnet 5 (claude-sonnet-5, Anthropic) reads the papers and writes each specification and its wording; code validates it and does all the arithmetic. Building and testing: Claude Code (Anthropic) with Claude Sonnet 5.5 (claude-sonnet-5-5). Audio overviews: scripts written with Claude Code; voice by Kokoro-82M, an open-source text-to-speech model run locally. Reviews: Claude Sonnet 5.5, Grok 4.5 (xAI), Gemini 3.5 (Google), GPT and GPT-5 Codex (OpenAI; exact builds not exposed). Interviewees, forum respondents and outside testers are described by role only.